

OMER FAROOQ

 (416) 844-5169  [Email](#)  [LinkedIn](#)  [GitHub](#)  [Portfolio](#)

EDUCATION

University of Waterloo

Expected: Apr 2028

Bachelor of Applied Science in Mechanical Engineering, Mechatronics Specialization

Waterloo, ON

- **Honors:** President's Scholarship of Distinction
- **Relevant Coursework:** Dynamics II, Thermodynamics I, Materials II, Circuits II, Mechanics of Deformable Solids II

EXPERIENCE

Manufacturing Engineering Intern

May 2025 – Aug 2025

ALMAG Aluminum

Brampton, ON

- Optimized part placement and equipment layout for **Tesla's Cybertruck** tonneau cover assembly, contributing to a **\$77M** manufacturing program focused on **lean production**, ergonomic flow, and scalable automation
- Executed a **Kaizen-driven** factory layout redesign, improving efficiency by **45%**, unlocking **38 days** of extra production per year, and generating **\$8.5M** in annual savings through 5S and streamlined flow
- Produced AutoCAD layouts for two facilities totaling **140,000 sq ft** and used them to design a **FIFO** flow for **70,000** parts, support **Value Stream Mapping**, and reduce internal travel distance by **27%** to boost part flow and efficiency
- Designed a CNC fixture in **SolidWorks** applying **GD&T** and tolerance stack-up analysis, minimizing setup time
- Created and updated **SOPs** for roll correction, **quality assurance (QA)**, and troubleshooting caustic system issues
- Built a **VBA-powered** tool to auto-highlight alloy substitutions from drop-downs, boosting material selection by **70%**

Design & Systems Engineering Intern

Sept 2024 – Dec 2024

Vision Extrusions Group

Toronto, ON

- Integrated a online database for **5,400+** dies, enabling **Just-in-Time (JIT)** access and improving efficiency by **72%**
- Performed **FEA** in **ANSYS Workbench** to optimize performance, enabling clamps to support **55+** kilograms
- Engineered an ultra-lightweight PVC-aluminum prototype window frame (**<2.3 kg**) using **DFM/A** in AutoCAD
- Achieved a **64%** increase in Moment of Inertia of a frame through material analysis and cross-sectional optimization
- Led a **Kaizen initiative** that decreased changeover downtime by **28%** and standardized workflows across extrusion lines

Mechanical Design Engineering Intern

Jan 2024 – Apr 2024

Vision Extrusions Group

Toronto, ON

- Engineered **8+ GD&T-based** drawings of window profiles in AutoCAD optimized for ease of **manufacturability**
- Used **Root Cause Failure Analysis** to enhance waterproofing by **22%**, ensuring an airtight seal against wind and rain
- Optimized a mass-produced window profile in AutoCAD to reduce PVC usage, yielding **\$26,000** in overall savings
- Applied **DFA** principles in AutoCAD to redesign window profiles for easier assembly and accuracy during fabrication

PROJECTS

Aerodynamics System | Python, Selenium, Google Cloud Console

January 2021

- Designed and programmed a **piano-playing robot** with accurate 3-axis movement, **3D-printed** parts, an ultrasonic sensor, and a color sensor to read and color-code sheet music, achieving **92%** accuracy while playing **70** notes

PianoBot | SolidWorks, C++, 3D Printing

Summer 2024

- Designed and programmed a **piano-playing robot** with accurate 3-axis movement, **3D-printed** parts, an ultrasonic sensor, and a color sensor to read color-coded sheet music, achieving an **92%** accuracy while playing **70** notes

Spin N' Shoot | SolidWorks, Laser Cutting, Arduino

Fall 2023

- Designed and programmed a **piano-playing robot** with accurate 3-axis movement, **3D-printed** parts, an ultrasonic sensor, and a color sensor to read and color-code sheet music, achieving **92%** accuracy while playing **70** notes

SKILLS

Tools: SolidWorks, AutoCAD, ANSYS Workbench, 3D Printing, Laser Cutting, CNC Milling, Extrusion Molding

Design & Manufacturing: FEA, GD&T, RCFA, DFM/A, JIT, 5S, Kaizen, Lean Six Sigma, Value Stream Mapping

Certifications: Certified SolidWorks Associate, Certified SolidWorks Professional, Microsoft Azure AI-900 & AZ-900

Programming Languages: C++, Excel VBA, Typescript, HTML5, SCSS, MATLAB